1 Direct Sums

If U, V, and W are subspaces of a vector space such that $W = U + V$, then every element $w \in W$ can be written as $w = u + v$ for some $u \in U$ and some $w \in V$. As we have seen, there might be many ways to write each w as the sum of an element of U and an element of V. However, it might also be the case that each element $w \in W$ can be written uniquely as the sum of an element of U and an element of V. In this case, we say that W is the *direct sum* of U and V.

Definition 1.1 (Direct sum)**.** Suppose V_1 and V_2 are subspaces of a vector space V. Then $V_1 + V_2$ is called a *direct sum* if each element of $V_1 + V_2$ can be written uniquely as $v_1 + v_2$ with $v_1 \in V_1$ and $v_2 \in V_2$. In this case, we write $V_1 \oplus V_2$ instead of $V_1 + V_2$.

More generally, if $V_1, \ldots, V_n$ are subspaces of V such that each element of $V_1 + \cdots + V_n$ can be written uniquely as $v_1 + \cdots + v_n$, where each $v_k \in V_k$, then we write $V_1 \oplus \cdots \oplus V_n$, or $\bigoplus_{k=1}^{n} V_k$ to denote that the sum is a direct sum.

Example 1.2.

(a) For the subspaces of F^3

$$U = \{(x,0,0) \in F^3\}, \quad V = \{(0,y,0) \in F^3\}, \quad W = \{(x,y,0) \in F^3\}$$

each element $(x,y,0)$ of W can be written uniquely as

$$(x,y,0) = (x,0,0) + (0,y,0),$$

so $W = U \oplus V$.

(b) For the subspaces of F^4

$$U = \{(x,x,y,y) \in F^4\}, \quad V = \{(x,x,x,y) \in F^3\}, \quad W = \{(x,x,y,z) \in F^4\}$$

each element $(x,x,y,z) \in W$ can be written as a sum of elements of U and V as

$$(x,x,y,z) = (-t+x-y+z, -t+x-y+z, -t+z, -t+z) + (t+y-z, t+y-z, t+y-z, t),$$

for any $t \in F$, so $W = U + V$, but this is not a direct sum.

(c) For each $k = 1, \ldots, n$, let $V_k = \{(0, \ldots, 0, \underbrace{x}_{k^{\text{th}}entry}, 0, \ldots 0) : x \in F\} \subseteq F^n$. Then it is straightforward to verify that

$$F^n = \bigoplus_{k=1}^{n} V_k$$

The definition of a direct sum requires that every vector in the sum to have a unique representation as an appropriate sum. The next result shows that when deciding whether a sum of subspaces is a direct sum, we only need to consider whether the zero vector can be uniquely written as an appropriate sum.

Proposition 1.3 (Condition for a direct sum)**.** *Suppose $V_1, \ldots, V_n$ are subspaces of a vector space V. Then $V_1 + \cdots + V_n$ is a direct sum if and only if the only way to write*

$$0 = v_1 + \cdots + v_n$$

is to take $v_1 = v_2 = \cdots = v_n = 0$.

Exercise 1.4. Prove Proposition 1.3.

The next result gives a simple condition for testing whether a sum of two subspaces is a direct sum.

Proposition 1.5. *Suppose U and W are subspaces of V. Then $U + W$ is a direct sum if and only if $U \cap W = \{0\}$.*

Exercise 1.6.

Prove Proposition 1.5. (*Hint*: Use Proposition 1.3.)

Remark 1.7. Note that Proposition 1.5 deals only with the case of two subspaces. When asking about a possible direct sum with more than two subspaces, it is not enough to test that each pair of the subspaces intersect only at 0. For example. one can check that

$$F^3 = V_1 + V_2 + V_3,$$

where

$$\begin{aligned} V_1 &= \{(x,y,0) : x,y \in F\}, \\ V_2 &= \{(0,0,z) : z \in F\}, \\ V_3 &= \{(0,y,y) : y \in F\}, \end{aligned}$$

is not a direct sum, yet $V_1 \cap V_2 = V_1 \cap V_3 = V_2 \cap V_3 = \{0\}$.

Exercise 1.8. Recall that a function $f : \mathbb{R} \to \mathbb{R}$ is *even* if $f(-x) = f(x)$ for all $x \in \mathbb{R}$ and *odd* if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$. Let V_e denote the set of all even functions and V_o the set of all odd functions. Show that $\mathbb{R}^{\mathbb{R}} = V_e \oplus V_o$.

Exercise 1.9. Let

$$U = \{(x,x,y,y) \in F^4 \mid x,y \in F\}.$$

Find a subspace W of F^4 other than $\{0\}$ such that $F^4 = U \oplus W$.